What the Meter Keeps

Interval Data Practices for Large Electricity Customers, and the Statistical Limits of Detecting Small Savings

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Abstract. This paper asks two questions that are usually treated separately. First, what electricity consumption data do utilities actually collect, retain, and make available for large commercial and industrial (C&I) customers? We document practices across 57 jurisdictions in North America, Europe, and the rest of the world, separating five layers that public discussion routinely conflates: the meter's measurement capability, the recorded interval, the transmission cadence, the resolution and history retained by the utility, and the data a customer or authorised third party can obtain. Second, given the data that realistically exists, how small a change in whole-building electricity consumption can modern statistical methods detect and quantify? Using the portfolio evidence on baseline-model accuracy and the ASHRAE Guideline 14 fractional-savings-uncertainty framework, we compute detection and quantification limits for savings of 2, 3, 5, 8, and 10 percent. Three findings stand out. Recording is rarely the binding constraint: large C&I meters record at 15 minutes across most of the European Union and much of North America, at 30 minutes across much of Asia-Pacific, and at 5 minutes in a small set of markets. Retention and access are the binding constraints: customer-obtainable history clusters at 12 to 24 months where it is documented at all, and utility retention schedules are unpublished in most of the jurisdictions studied. Statistically, daily or hourly interval data are sufficient to quantify savings of about 5 percent at a typical well-modelled site, and about 3 percent at 68 percent confidence; savings of 2 percent are detectable but generally not quantifiable at an arbitrary site; and monthly billing data cannot support any of these claims. Finer-than-hourly data adds little detection power once residual autocorrelation is counted, because the binding constraint is unexplained operational variance at the scale of days to seasons, not sampling density.

Keywords: interval data; advanced metering infrastructure; measurement and verification; baseline models; savings uncertainty; data access; electricity markets.

JEL codes: Q41, L94, C52, L51.

Non-technical summary. Every large factory, office tower, hospital, and data centre has a main electricity meter, and in most rich countries that meter quietly records consumption every 15 or 30 minutes. This paper checks what happens to those recordings: how long the utility keeps them, and whether the customer can get them back. The short answer is that the recordings usually exist, the right to obtain them exists in a growing list of jurisdictions, but the obtainable history is often short (one to two years), and in most countries nobody publishes how long the utility itself keeps the data. The paper's second half asks what such data can prove. If a building cuts its electricity use by a few percent, can statistics separate that cut from normal variation in weather, schedules, and operations? The answer is a qualified yes. With a year of daily or hourly data before and after, a 5 percent reduction can be measured tightly enough to act on at a typical site, and a 3 percent reduction can be measured at moderate confidence. A 2 percent reduction sits at the edge: visible, but not precisely measurable at an ordinary site. Monthly bills alone prove almost nothing at these sizes. And collecting data every minute instead of every 15 minutes does not fix the problem, because

the noise that matters is not in the minutes; it is in the building's changing operations from week to week.

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1 Introduction

Two facts about large electricity customers are both true and rarely stated together. The first is that their consumption is measured at fine temporal resolution almost everywhere: interval metering is a precondition of time-of-use tariffs, demand charges, and wholesale market participation, and it long predates residential smart-meter programmes. The second is that almost everything downstream of the measurement (how long the record is kept, at what resolution it is stored, and whether the customer can retrieve it) is set by a patchwork of market rules, metering codes, and utility practice that has never been mapped in one place.

The mapping matters because the uses are concrete. Energy managers, efficiency programmes, and demand-management services all promise to reduce a site's consumption by single-digit percentages and to prove the reduction from meter data. Whether that proof is possible depends jointly on two independent questions: an institutional question (does the historical record exist, and can the customer obtain it?) and a statistical question (given the record, how small a change is distinguishable from the building's own variability?). A failure on either side voids the exercise. Data that exists but cannot be obtained is useless; data that can be obtained but cannot support the inference is equally useless; and the two failures have different remedies, which is why this paper treats them in sequence and joins them only at the end.

The paper makes four contributions. First, it documents interval-data practices for large C&I customers across 57 jurisdictions, organised by a five-layer distinction (measurement, recording, trans-

mission, retention, access) that prevents the most common error in this literature: treating a settlement interval, a meter capability, and a customer right as the same thing. Second, it identifies where the evidence base is thin (retention schedules above all) and says so, rather than filling gaps with generalities. Third, it assembles the empirical distribution of whole-building baseline-model accuracy from the portfolio literature and applies the standard savings-uncertainty mathematics to compute, transparently, what savings of 2 to 10 percent require in data, duration, and model quality. Fourth, it joins the two halves: it states which of the documented data regimes are statistically sufficient for which sizes of claim.

The headline results are as follows. On the institutional side, recorded resolution for large C&I main meters is 15 minutes across the European Union (by settlement design) and in the best-documented North American markets, 30 minutes across much of Asia-Pacific, and 5 minutes in a minority of markets (Australia's National Electricity Market, the Philippines, Ontario wholesale metering, and one documented US retail case). Customer-obtainable history, where documented, clusters at 12 to 24 months. Utility retention is documented in only a handful of jurisdictions (Australia's seven-year metering-data rule is the cleanest case) and is otherwise unpublished: this is the single largest evidence gap in the study. On the statistical side, the binding quantities are the baseline model's out-of-sample error and the effective (autocorrelation-adjusted) number of observations. Monthly billing models cannot detect savings below roughly 4 to 6 percent even at 68 percent confidence, and cannot quantify savings below roughly 8 to 14 percent. Daily-aggregated interval models at the median portfolio building quantify 5 percent savings at 90 percent confidence and 3 percent at 68 percent confidence over 12-month baseline and reporting periods. Moving from hourly to 15-minute or 1-minute data changes these thresholds by at most about a quarter, and by nothing at 1 minute, once residual autocorrelation is counted.

The rest of the paper proceeds as follows. Section 2 sets out the five-layer framework and the institutional mechanics. Section 3 documents practices by region. Section 4 constructs a graded global coverage estimate. Section 5 develops the statistical assessment. Section 6 joins the two halves. Section 7 states what evidence would change the conclusions. Section 8 concludes.

2 Institutional background: five layers, not one

Public discussion of "smart meter data" usually treats resolution as a single attribute of a meter. For large C&I customers, five distinct layers determine what data exist and what can be done with them, and the layers are set by different instruments and different actors.

Layer 1: measurement. The meter's physical capability: the integration or sampling process by which energy is accumulated. Modern revenue meters measure continuously and can typically form intervals far finer than any market requires; German metrology rules, for example, permit 5-minute measurement periods at the device (Stream 2, Section 5.4 of this paper's evidence base). Layer 1 is almost never the constraint for large C&I sites.

Layer 2: recording. The interval at which consumption is written to the meter's load-profile memory and, downstream, the interval of the record the utility's metering system holds. This is the layer that market design usually fixes: settlement-grade recording is 15 minutes in the EU and in Texas, 5 minutes in Australia, 30 minutes in Japan and Singapore. Recording can be finer than settlement (Brazil's free-market meters record 5-minute mass-memory data that settlement consolidates to hourly) and settlement can be finer than recording (PJM and ISO New England derive 5-minute settlement quantities from hourly meter data by profiling; the fine series in the settlement system is manufactured, not measured).

Layer 3: transmission. How often the recorded data move from meter to utility. Transmission is typically a daily batch even where recording is 15-minute (France's Linky system; Austria's smart-meter configurations), which matters for operational uses but not for historical analysis.

Layer 4: retention. How long the utility, market operator, or data hub keeps the interval record. Retention is set by metering codes, market rules, data-protection law, and internal practice, and it is the least visible layer: most jurisdictions in this study publish no retention schedule for interval data at all.

Layer 5: access. What the customer, or a third party with the customer's authorisation, can obtain: which resolution, how much history, through what mechanism (portal, API, standardised data-sharing framework such as Green Button Connect or a consumer data right, formal request, or paid service), and at what cost. Access is a legal and commercial construct, not a technical one; the same record can be a statutory right in one jurisdiction (the EU, Ontario, Australia) and a paid product in another (Hydro-Québec's tariffed load-profile services).

Two structural points follow. First, the layers can diverge in either direction, and both directions are documented in Section 3: utilities that record more than customers can see (the general pattern found by the EIA-commissioned assessment of US practice), and settlement systems that display more granularity than was ever measured (PJM load). Any study that treats a retrieved series as raw measurement without establishing which layer it came from risks analysing a profile artefact. Second, for the statistical question of Section 5, the operative layers are 4 and 5 jointly with 2: a baseline model needs recorded-resolution data (Layer 2), over a baseline period that retention must cover (Layer 4), through a channel the analyst can actually use (Layer 5).

Customer classification mediates every layer. Large C&I customers are typically defined by demand or consumption thresholds (for interval metering mandates, market participation, or data rights) that vary by jurisdiction; this paper uses "large C&I" functionally, meaning customers on interval metering by rule or by tariff, and notes thresholds where they drive the data regime (for example, Australia's Consumer Data Right, which excludes customers above 5 GWh per year and so routes the largest users to a different access mechanism than the statutory API right).

3 Documented practices by region

This section summarises the evidence base. Every practice stated here is documented in the underlying research files with source title, locator, and retrieval date (16 October 2026), and carries an evidence grade (Observed, Reconstructed, Modelled, or Assumed, at quality A, B, or C). Where documentation could not be found after genuine retrieval attempts, the paper records the field as Unknown. Appendix A reproduces the jurisdiction tables in full.

3.1 North America

The documented North American standard for large C&I recording is 15 minutes: Texas interval data recorders under the Public Utility Commission of Texas rule, California retail C&I metering, and the commercial portals of ComEd and Duke Energy all sit at 15 minutes. Two verified finer cases exist: Ontario wholesale revenue metering collates at the 5-minute dispatch interval (IESO Market Rules), and Con Edison delivers 5-minute data to commercial AMI customers through Green Button Connect, the finest customer-obtainable resolution documented anywhere in this study. Against that, US wholesale load metering for settlement remains hourly in NYISO and PJM, and FERC Order No. 825, which mandated 5-minute real-time settlement, expressly did not require 5-minute revenue

metering; PJM manufactures its 5-minute load settlement quantities from hourly data. CAISO's tariff keys the submission interval to meter capability (5 minutes mandatory for ancillary-service resources; 5, 15, or 60 minutes electable otherwise).

Customer-obtainable history clusters tightly at 12 to 24 months: PG&E's Rule 22 provides 12 months to authorised third parties; Con Edison documents a hard 2-year cap; Duke Ohio's EDI channel caps aggregator history at 12 months. Illinois is the outlier on the duration of ongoing sharing (nonresidential Green Button Connect sharing can continue indefinitely until rescinded under ICC 17-0123), which is a different quantity from depth of history. Internal retention is the region's largest evidence gap: only two hard figures were found (CAISO's duty on scheduling coordinators to retain settlement-quality meter data for at least 3 years for DER aggregations, and Hydro-Québec's demonstrated possession of at least 4 years of hourly data, released to researchers under agreement). For ERCOT's distribution companies, PJM, NYISO, ComEd, Duke, IESO, Ontario's metering repository, and BC Hydro, no public retention schedule was found, and this paper does not estimate one.

Canada's three documented regimes differ sharply. Ontario mandates Green Button by regulation (O. Reg. 633/21, in force 1 November 2023) over an hourly distribution metering repository; the mandated history depth was not verified against the regulation text. BC Hydro's documented customer product is hourly (MyHydro), with large-C&I recording and retention undocumented. Québec has no access right of the Ontario type; load profiles are a paid, tariffed service (Visilec, VigieLigne), an arrangement in which granular data is a product rather than an entitlement.

3.2 Europe

Europe is the one region where recording resolution is set by supranational design. The Electricity Balancing Guideline fixes a 15-minute imbalance settlement period; Directive (EU) 2019/944, Article 20(g), ties smart-meter settlement to that resolution; and the single day-ahead coupling moved to 15-minute products for delivery from 1 October 2025. National verification in this study confirmed 15-minute recording for large C&I in Denmark (Executive Order 477/2023), Sweden (from November 2023), Germany (a 15-minute floor for billing-relevant registration under the PTB-A 50.8 metrology rules), and Austria (15-minute values in the extended meter configuration). The United Kingdom is the major exception: Great Britain settles C&I half-hourly (the P272 reform, now migrating under Market-wide Half-Hourly Settlement to May 2027).

Retention is where European harmonisation stops. No EU-wide retention period for interval data exists. The documented points are few: Denmark's DataHub requires market actors to document message exchanges for at least 3 years and erases personal data after 5 years; in Great Britain the retrievable record through the Data Communications Company is meter-resident (13 months of half-hourly data, 2 years of daily data, per the government's business guide); Austria holds 60 days on the meter, with the distributor archive beyond that unverified. Statutory interval-data retention for Germany, France, the Netherlands, Spain, Italy, and Poland could not be established from primary sources in this study and is recorded as Unknown.

Access, by contrast, is a legal right across the EU: Article 20 of the Directive guarantees free access to validated historical data and free delegation to third parties, and Implementing Regulation (EU) 2023/1162 harmonises the role model (metering data administrator, data access provider) without harmonising history depth. Working routes include hub portals (Denmark's Eloverblik; Poland's CSIRE, live from 2 July 2025 and free of charge), distributor and supplier portals, and contracted metering companies for large users in the Netherlands. Documented internal-versus-visible gaps include Austria (daily totals by default, 15-minute values on opt-in, with the distributor permitted

to use anonymised 15-minute data internally regardless), Germany (5-minute measurement at the device, 15-minute billing record), Great Britain (half-hourly settlement against a 13-month meter-resident retrieval window), and Norway (a hub ready for 15-minute data from September 2023 while settlement consumed hourly values under a derogation to April 2024).

3.3 Rest of the world

Fourteen jurisdictions were documented in the main pass across Asia-Pacific, Latin America, the Middle East, and Africa, and a supplementary pass added seventeen more. Settlement and recording granularity: 5 minutes in Australia (since October 2021) and the Philippines; 15 minutes in China's spot-market provinces (96-point curves), in South Korean AMI load profiles, and in India's 15-minute deviation-settlement blocks with 0.2S-class interface meters; 30 minutes in New Zealand, Singapore, Japan, Vietnam's wholesale market, and South Africa's Megaflex time-of-use metering; hourly balances in Chile. Brazil's free market holds the clearest documented case of recording finer than settlement: meters keep 5-minute mass-memory data, collected daily and consolidated to hourly for settlement, and market agents can access the 5-minute record itself.

Australia is the only jurisdiction in this stream with a clean, citable, system-wide retention rule: metering data providers must keep metering data and audit trails for at least 7 years. New Zealand's forthcoming consumer data designation (from 1 July 2027, half-hourly data, 2-year look-back) implies retrievable history of at least 2 years. Everywhere else in the stream, retention is Unknown. Rights-based access is similarly concentrated: Australia's Consumer Data Right for energy (with eligibility capped below 5 GWh per year, so the largest customers use the meter-data request route instead), New Zealand's statutory designation, India's statutory duty to read smart meters remotely at least daily and share data with consumers (a coarse granularity), and South Korea's consent-based Energy MyData (household-framed at launch; C&I coverage undocumented). Elsewhere, access runs through market rules or bilateral utility relationships. Interval metering for large C&I is effectively universal across these jurisdictions (it is a condition of time-of-use tariffs and market participation and long predates mass smart-meter rollouts), so the constraints are retention and access, not the meter; Dubai and Saudi Arabia have near-universal smart fleets across all customer classes, with the customer-facing granularity and any interval data right undocumented for Saudi Arabia and unspecified for Dubai.

The supplementary seventeen sit mostly in single-buyer systems with monthly billing, and their documented recording falls in the same narrow band as the main pass. At 30 minutes: Pakistan, where the DISCO AMI specification for CT-operated large-consumer meters sets 30-minute sampling with twice-daily retrieval, configurable down to 1 minute; Bangladesh, whose standard meter specification requires a 30-minute load survey; Malaysia, where maximum demand is defined on a 30-minute block and smart meters record and transmit every 30 minutes; and Nigeria, whose Metering Code requires maximum-demand meters to store demand in every 30-minute period. At one hour, where the system is settlement-led or AMR-led rather than AMI-led: Türkiye, with hourly settlement under the EPİAŞ Balancing and Settlement Regulation and settlement meters read through the OSOS system; Colombia, with hourly frontier reporting to the ASIC under the CREG metering code and estimates barred from settlement formats since 2021; Oman, where a central AMR system covering roughly 12,000 accounts above 150 MWh per year delivers hourly readings; and Abu Dhabi's settlement points, which store hourly demand values at metering outstations. Argentina is the supplement's 15-minute case: large users above 1 MW must install remotely readable SMEC metering, and market data at the wholesale interface are formed in 15-minute records (grade B, retained from the working research on CAMMESA's technical procedure PT-4; current regulation separately confirms 15-minute

demand records at that interface). Sri Lanka's documented 15-minute recording is residential only (LECO smart meters in a peer-reviewed dataset descriptor) and is not evidence of C&I practice, which is Unknown there. For Nepal, Indonesia, Thailand, Qatar, Israel, Egypt, and Kenya, deployment is documented but the production recording interval is Unknown in accessible sources: Nepal's time-of-day meters are the billing record of consequence (a multi-year arrears dispute was settled on those records); Indonesia read a Bali AMI pilot every 30 minutes and runs AMI at scale in Jakarta; Thailand identified automatic reading for its 400-plus largest C&I revenue meters and deferred it; Qatar has AMI at a scale above 500,000 meters; Israel is rolling smart meters out to full coverage by 2028; Egypt is replacing a 28.8 million meter stock with prepaid meters alongside a 250,000-meter smart pilot; and Kenya procures smart metering specifically for large power customers.

Retention in the supplement follows the main pass's pattern, with one refinement: where a number exists at all, it sits inside a device or outstation specification rather than in published system-level policy. Bangladesh's meter specification requires at least 90 days of load survey and 12 months of energy and demand data held in the meter. Nigeria's Metering Code requires six months of half-hourly demand and energy history in the meter. Abu Dhabi's Metering and Data Exchange Code requires at least 40 days of hourly demand values at settlement outstations. System-level retention is Unknown in 14 of the 17; Türkiye, the nearest exception, assigns its transmission operator a duty to store settlement measurement data but states no period. Access is correspondingly thin. Malaysia is the supplement's one clear case of the customer routinely seeing the recorded interval: smart-meter customers view consumption at up to 30-minute resolution in the myTNB portal, though portal history depth is undocumented. Bangladesh's DPDC AMI tender contractually requires the meter data system to give customers current and historical interval data through a portal, but the live product is unverified. Nepal's industrial customers received their own time-of-day records only as utility-prepared spreadsheet summaries, and only under dispute pressure. In Argentina, Colombia, and Türkiye, interval data flows to market agents and participants rather than to end customers as a documented right, and no general customer interval right was found in Türkiye. The documented internal-finer-than-visible gaps are correspondingly specific: Abu Dhabi customer meters must integrate demand over periods programmable down to 5 minutes while the documented customer product is a billing application of unstated granularity; Pakistan's large-consumer meters are configurable to 1-minute sampling against no documented customer product at all; and Nigeria's meter holds a six-month half-hourly history of which the customer is documented to receive one maximum-demand number per month.

4 A global coverage estimate, with its basis stated

The commissioning question asks what proportion of large C&I sites globally have access to historical data at each resolution. No measured global statistic exists, and this paper does not manufacture one. What the evidence supports is a structured judgement over the 57 documented jurisdictions, which include most of global C&I load (China, the United States, the EU, India, Japan, Brazil, and Australia are all in the sample). The estimates below are graded Modelled-C throughout: they are the authors' synthesis of the case evidence, stated with ranges, and the falsifiability section specifies what would replace them with measurements. The seventeen supplementary jurisdictions of Section 3.3 do not disturb these estimates: they are predominantly smaller markets, and where they document retention at all they document it in-device rather than at system level, so the load-weighted shares stand on the enlarged sample.

Two readings of Table 2 matter for Section 5. First, the data regime a savings claim can realistically assume, in most of the developed world, is 15 or 30-minute data with 12 to 24 months of obtainable

Table 1: Recorded resolution and customer-obtainable history, large C&I main meters, selected jurisdictions (documented practice; see Appendix A for the full tables and grades)

Jurisdiction	Recorded	Obtainable history	Access basis
Texas (ERCOT)	15 min	∼12–36 mo	Joint TDSP portal
California	15 min retail	12 mo (third party)	Green Button Connect (CPUC rule)
New York (Con Ed)	5 min (AMI)	24 mo cap	Green Button Connect (NYPSC framework)
Illinois (ComEd)	15 min	12–24 mo; sharing indefinite	Green Button Connect (ICC order)
Ontario	hourly (dist.); 5 min (wholesale)	depth unverified	Green Button mandated (O. Reg. 633/21)
EU (framework)	15 min	depth set nationally	Statutory right (Dir. 2019/944 Art. 20)
Germany	15 min	depth unknown	Supplier/operator provision
Great Britain	30 min	13 mo half-hourly (DCC)	SEC consent route; DCC retrieval
Denmark	15 min	depth unknown	Hub portal (DataHub/Eloverblik)
Australia (NEM)	5 min	multi-year (7-yr retention)	Data request; CDR (<5 GWh/yr)
New Zealand	30 min	2 yr (from Jul 2027)	Statutory data right (designation)
Singapore	30 min	via retailer	Market rules (MSSL)
Japan	30 min	via retailer/T&D	Market framework
South Korea	15 min	portal; MyData (scope unclear)	Utility portal; consent regime
India	15 min	on request (discom)	Statutory duty (coarse)
China (spot provinces)	15 min	via trading centre/retailer	Market rules
Brazil (free market)	5 min recorded	via CCEE agent access	Market rules (SCDE)
South Africa	30 min	bilateral	Utility account channels
Chile	hourly	bilateral	Market/utility practice
Malaysia	30 min	up to 30 min (portal); depth undocumented	myTNB portal (utility product)
Pakistan	30 min (AMI spec)	no documented product	None documented
Bangladesh	30 min (spec)	portal access specified (DPDC tender); live status unverified	DPDC AMI requirement
Türkiye	hourly	participant data; general right not found	Market systems (participants)
Argentina	15 min (grade B)	agent data; depth Unknown	CAMMESA agent channels
Nigeria	30 min (in-meter MD)	monthly MD figure only	No documented right
Abu Dhabi	hourly (settlement); 5 min capable (customer meters)	app view; granularity Unknown	TAQA Distribution app
Colombia	hourly (frontier)	agent channels; end-customer right Unknown	Market rules (ASIC)
Oman	hourly (AMR)	not documented	None documented

Table 2: Estimated share of large C&I sites, documented-jurisdiction sample (Modelled-C; structured judgement from Section 3 evidence, not a measured statistic)

Quantity	Estimate	Basis
Site is interval-metered at 30 min or finer	85–95%	Interval metering is a condition of TOU tariffs and market participation in nearly all sampled jurisdictions
A 15-min (or finer) series is recorded	55–70%	EU, most documented US markets, Australia, Brazil, Korea, China spot provinces vs. 30-min Asia-Pacific cluster and hourly wholesale legacies
A defined channel exists to obtain $\geq$12 months of history at recorded resolution	45–60%	Rights-based regimes (EU, parts of NA, AU, NZ) plus market-rule channels; bilateral-only regimes counted at half weight
Documented retention of $\geq$2 years can be verified	15–25%	Australia (7 yr), Denmark hub rules, UK meter-resident 13 mo (below 2 yr at HH), Hydro-Québec (demonstrated, not published)
Customer can obtain 5-min or finer history	5–10%	Con Edison, Ontario wholesale, CAISO AS loads, Australia, Brazil agents
Customer can obtain 1-min history from utility records	<2%	No documented utility channel at 1 min in any sampled jurisdiction

history. Second, the constraint binds in a specific order: recording is ample, obtainable history is adequate for a 12-month baseline in roughly half of jurisdictions (and obtainable by negotiation in more), and verified long retention is rare. A research or commercial design that requires more than 24 months of sub-hourly utility history, or any 1-minute utility data, is not supported by documented practice anywhere in the sample.

5 How small a saving can the data prove?

5.1 The measurement problem

Whole-building measurement and verification (IPMVP Option C) estimates savings as the difference between a baseline model's prediction of what the building would have used in the reporting period and what it actually used. Model error is therefore savings error, one-for-one. The two standard error metrics, per ASHRAE Guideline 14, are the coefficient of variation of the root mean square error, CV(RMSE), which measures the typical size of the per-period prediction error relative to mean consumption, and the net mean bias error, NMBE, which measures systematic bias. Both are defined per data period, so a CV(RMSE) is meaningless without its timescale.

The decisive evidence is out-of-sample: models trained on one period and scored predicting a later one, which is the correct analogue of verification and is systematically worse than in-sample fit. Three portfolio studies from Lawrence Berkeley National Laboratory provide the distributions (Table 3). The median commercial building is predictable to about ±13 percent per day or ±17 to 20 percent per hour or 15-minute interval, out of sample. The best quartile of buildings reaches about 8.6 percent daily CV(RMSE); the worst quartile exceeds 23 to 34 percent. Model class matters less than the building: gradient boosting beat the time-of-week-and-temperature regression in about 81 percent of buildings but by only about 2 points of CV(RMSE) at the median (17.4 vs. 19.3 percent). Median bias is small (NMBE of −0.3 to −1.2 percent) but the central half of buildings spans roughly −5 to +5 percent, so uncorrected bias alone can swamp a 2 to 3 percent savings claim. Shorter baselines cost accuracy: cutting training from 12 to 3 months raises the median daily CV(RMSE) from 12.9 to 20.5 percent.

Table 3: Out-of-sample CV(RMSE) percentiles across buildings (Observed-A: Granderson et al. 2016; Touzani et al. 2018)

Study / data	Model class	Timescale	25th	Median	75th
537 buildings, 10 models	mixed	Daily totals	8.6%	12.9%	23.5%
537 buildings, 10 models	mixed	15-min totals	13.5%	19.7%	33.5%
410 buildings	Gradient boosting	Hourly model	12.4%	17.4%	27.2%
410 buildings	Time-of-week & temp.	Hourly model	13.6%	19.3%	29.6%
410 buildings	Random forest	Hourly model	13.6%	19.5%	33.6%

Two calibration facts discipline everything that follows. First, the empirical null distribution: applying standard models to 400 buildings with no intervention, the apparent "savings" fell within ±5 percentage points in 50 percent of cases and within ±20 points in 90 percent (Granderson et al. 2015). That is the noise floor for an unscreened site before any formula is applied. Second, a variance decomposition (Granderson and Price 2014) shows why the ceiling exists: temperature explains only 5 to 15 percent of hourly load variance; the weekly schedule explains much of the rest for regular buildings; and an unexplained component (occupancy, behaviour, equipment and activity change) is under 15 percent of variance for most buildings but over 50 percent for a few. No algorithm

can capture a component for which no explanatory variable exists. Applied to utility programme data, automated whole-building verification found about 70 percent of buildings well suited; the complementary 30 percent are where small-savings claims fail.

ASHRAE Guideline 14's widely quoted tolerances (monthly NMBE within ±5 percent and CV(RMSE) at most 15 percent; hourly ±10 percent and 30 percent) are per-period calibration criteria, conventionally evaluated in-sample. They are not savings-accuracy guarantees: Section 5.3 shows a model exactly at the monthly tolerance leaving a 5 percent saving unquantifiable over any practical horizon.

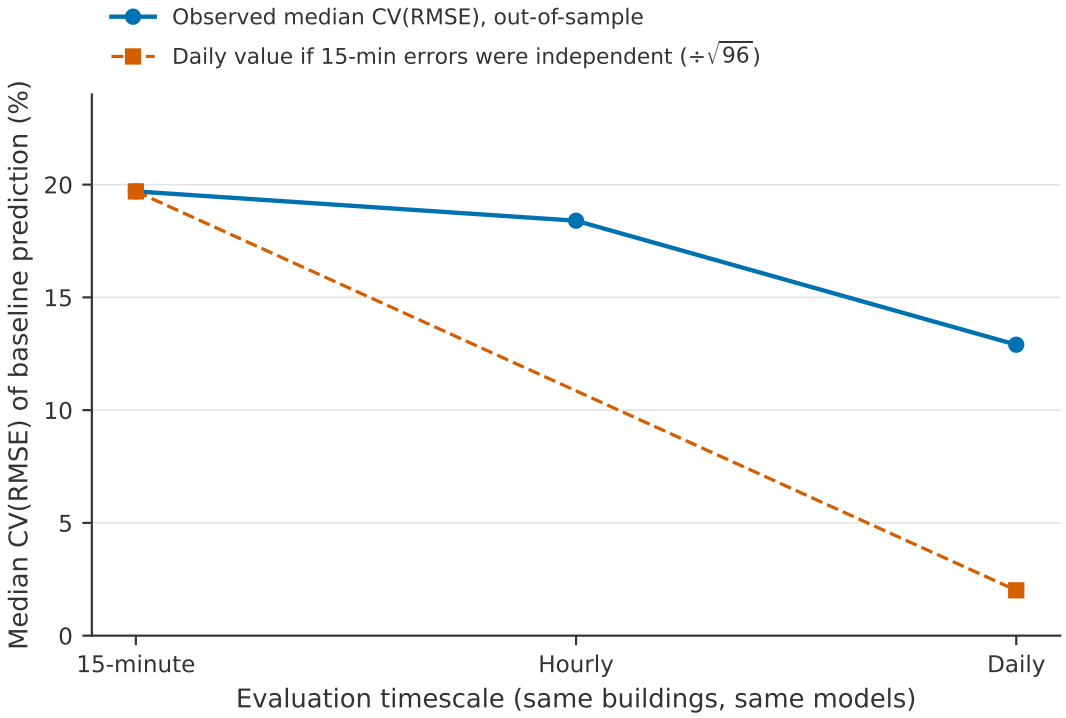


Figure 1: Median out-of-sample CV(RMSE) by evaluation timescale for the same buildings and models (Granderson et al. 2016; hourly value averages the gradient-boosting and time-of-week-and-temperature medians of Touzani et al. 2018), against the daily value that independent 15-minute errors would imply. Aggregation from 15 minutes to daily reduces error by a factor of 1.53, not the $\sqrt{96} \approx 9.8$ that independence would give: most of the error is a persistent day-scale component.

5.2 The uncertainty framework

Let the baseline model be fitted on n periods and applied over m reporting periods, with CV the model's CV(RMSE) at the working timescale and F the savings fraction. If per-period errors were independent, the fractional savings uncertainty (FSU) at critical value t would be

$$\text{FSU} = \frac{t \cdot \text{CV}}{F} \cdot \sqrt{\frac{1}{m} + \frac{1}{n}},$$

which exposes the scaling laws: FSU is linear in model error, inversely proportional to the savings fraction, and improves only with the square root of the number of periods. Guideline 14's Annex B form, as implemented in the CalTRACK methods used by pay-for-performance programmes, adds a

parameter-uncertainty constant and, critically, an autocorrelation correction:

$$\mathrm{FSU} \;=\; \frac{t \;\cdot\; 1.26 \;\cdot\; \mathrm{CV(RMSE)}}{F}\,\sqrt{\frac{n}{n'}\cdot\frac{1+2/n}{m}}, \qquad n' \;=\; n\,\frac{1-\rho}{1+\rho},$$

where ρ is the lag-1 autocorrelation of the baseline residuals and n' is the effective number of independent baseline observations. Ignoring autocorrelation understates FSU by the factor $\sqrt{(1+\rho)/(1-\rho)}$: 1.36 times at $\rho = 0.3$, 2.17 times at $\rho = 0.65$, 4.36 times at $\rho = 0.9$, and 14 times at $\rho = 0.99$. Sub-daily building-load residuals are strongly serially correlated, so this term is not a refinement; it is the difference between honest and fictitious precision (Walter, Price and Sohn 2014; Touzani et al. 2018, on the blocked cross-validation that honest error estimation requires).

Practice uses two criteria. Detection means the confidence interval for savings excludes zero (FSU at most 100 percent). Quantification, the standard a payment or published claim should meet, is FSU at most 50 percent at 68 percent confidence (FEMP and Guideline 14 Annex B convention), with at most 25 percent graded good in LBNL's M&V 2.0 guidance at 80 to 90 percent confidence. A stricter hypothesis-testing framing (90 percent confidence with 80 percent power) would raise every threshold in Section 5.3 by roughly half again; the paper uses the M&V convention, as is standard in this literature, and says so.

5.3 Detection and quantification limits for 2 to 10 percent savings

Table 4 defines five scenarios spanning the data regimes of Section 3, and Table 5 gives the resulting limits, computed from the Guideline 14 formula (Modelled; inputs Observed as marked). The results answer the commissioning question directly.

Table 4: Scenarios for the threshold computations (CV(RMSE) inputs Observed-A from Table 3; autocorrelation inputs Assumed where marked)

ID	Model and timescale	CV	Baseline n	Reporting m	ρ
A	Monthly billing, good quality	10%	12 mo	12 mo	0
B	Monthly billing, Guideline 14 tolerance	15%	12 mo	12 mo	0
C	Daily totals, median building	13%	365 d	365 d	0.3 (Assumed)
D	Daily totals, top-quartile building	8.5%	365 d	365 d	0.3 (Assumed)
E	Daily totals, median, short reporting	13%	365 d	90 d	0.3 (Assumed)

Table 5: Minimum detectable (FSU $\leq$ 100%) and quantifiable (FSU $\leq$ 50%) savings, percent of consumption (Modelled from the Guideline 14 formula)

Scenario	Detect 68%	Detect 90%	Quantify 68%	Quantify 90%
A: monthly, CV 10%	4.1	7.1	8.3	14.2
B: monthly, CV 15%	6.2	10.7	12.4	21.3
C: daily, median	1.2	1.9	2.3	3.9
D: daily, top quartile	0.8	1.3	1.5	2.5
E: daily, 90-day report	2.4	3.9	4.7	7.8

Read against the target savings of 2, 3, 5, 8, and 10 percent:

2 percent. Detectable at a median building with daily data and 12+12 months (FSU 59 percent at 68 percent confidence, 96 percent at 90 percent), but quantifiable at neither level; quantifiable at 68 percent only at a top-quartile building. From monthly data, or with a 90-day reporting period, a 2

percent claim fails even detection. A blanket claim that savings below 10 percent are undetectable is false; a claim that 2 percent is routinely provable is also false.

3 percent. Quantifiable at 68 percent and detectable at 90 percent at the median daily-data building; quantifiable at both levels at a top-quartile building. Not supportable from monthly billing data at any practical duration.

5 percent. Quantifiable at both confidence levels at the median daily-data building (FSU 23 and 39 percent) and comfortably at a top-quartile building. From a good monthly model it is detectable only at 68 percent after a full year; at the Guideline 14 tolerance CV of 15 percent, quantifying 5 percent at 68 percent would require about 74 months of reporting, an operationally meaningless duration.

8 and 10 percent. Quantifiable under every daily-data scenario at both levels, and at 10 percent quantifiable at 68 percent even from a good monthly model. These are the sizes at which single-site whole-building claims become robust across data regimes.

Reporting duration is the practitioner's main lever after model quality: shortening reporting from 12 months to 90 days roughly doubles every FSU (Scenario E), while the square-root law means extending beyond about 12 months buys little. Two cautions complete the picture. The formula gives the floor of uncertainty under a correctly specified, stationary baseline; the empirical null distribution of Section 5.1 (apparent savings within ±20 points in 90 percent of unscreened buildings) is the realistic scale when operational drift is present, and both must be reported. And portfolio aggregation changes the level of analysis entirely: portfolios of about 40 buildings achieved 1.5 to 4 percent total prediction error, which is why programme-level savings of a few percent are routinely defensible while site-level claims of the same size are not.

5.4 Does finer resolution help?

The empirical comparison is direct, because the same studies scored the same buildings at several timescales. Aggregating 15-minute data to daily totals reduces the median CV(RMSE) from 19.7 to 12.9 percent, a factor of 1.53. Had the within-day errors been independent, the factor would have been $\sqrt{96} \approx 9.8$. The shortfall is the autocorrelation: most prediction error is a persistent day-scale component (the day's schedule level, that day's temperature response, an unmodelled operational state) that no amount of within-day sampling cancels. Table 6 computes the consequence for detection at a 5 percent saving: moving from monthly to daily evaluation cuts FSU roughly three-fold (79 to 23 percent at 68 percent confidence), because it multiplies the effectively independent observations and lets the model resolve schedule and temperature response. Moving from daily evaluation to hourly or 15-minute evaluation buys a further 20 to 25 percent at most. Moving to 1-minute evaluation buys nothing in the calculation, because per-interval error rises and lag-1 autocorrelation approaches 1 fast enough to offset the larger raw count. Setting $\rho = 0$, by contrast, would make 15-minute data look about 6 times better than it is and 1-minute data about 14 times better; any claim of 1 to 2 percent detection from sub-hourly data that rests on interval counts alone should be presumed to have made that error.

Finer data does have real benefits, of a different kind: characterising temperature response and schedule from shorter baselines; locating when consumption changed (change-point timing and non-routine-event diagnosis); demand (kW) analysis; and fault detection. These are diagnostic benefits. For cumulative-savings detection at a site, the binding constraint is low-frequency unexplained variance, which sampling density cannot reduce. Availability and usefulness therefore cut both ways: a utility retaining only hourly data has not thereby disabled rigorous small-savings verification (daily

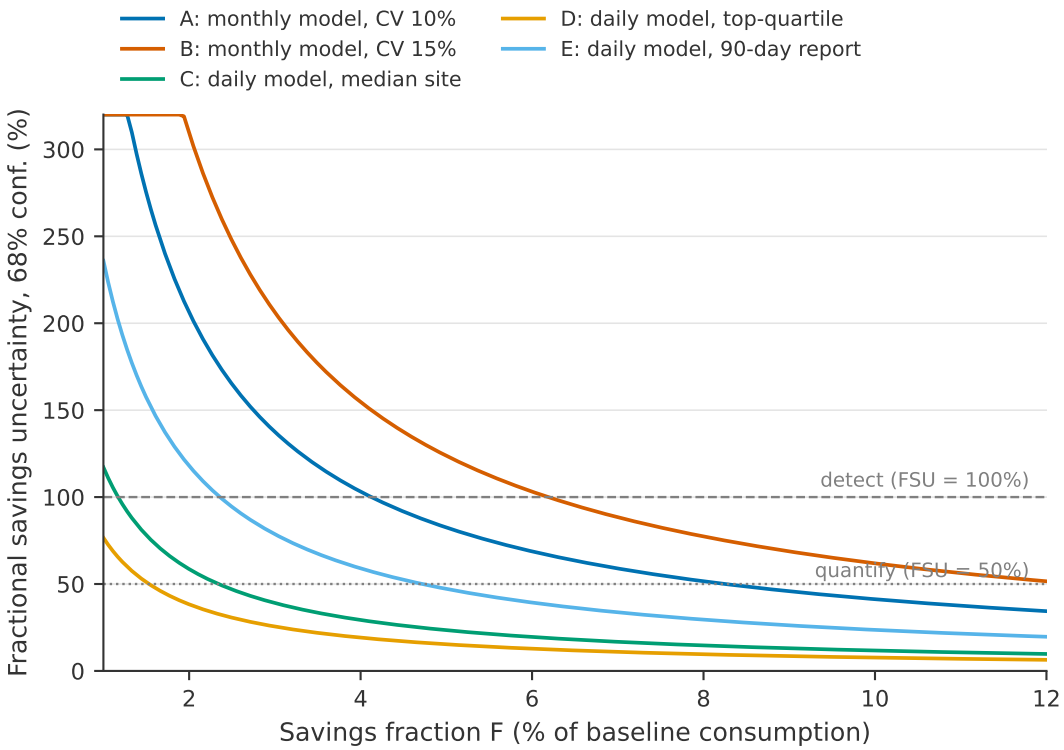


Figure 2: Fractional savings uncertainty at 68 percent confidence against the savings fraction, by scenario (Guideline 14 formula; Table 4 inputs). The horizontal lines mark the detection (100 percent) and quantification (50 percent) criteria. Monthly-data scenarios (A, B) sit above the quantification line across the whole 2 to 10 percent range; daily-data scenarios (C, D) cross it at small fractions.

Table 6: FSU at a 5 percent saving, 68 percent confidence, 12-month baseline and reporting, by evaluation timescale (Modelled; CV inputs Observed where marked, sub-daily ρ and the 1-minute CV Assumed; no portfolio study of 1-minute verification data was located)

Timescale	CV(RMSE)	Raw n (1 yr)	ρ	Effective n'	FSU (naive ρ=0)
Monthly totals	10% (Obs.)	12	0	12	79% (79%)
Daily totals	12.9% (Obs.)	365	0.3	197	23% (17%)
Hourly	19% (Obs.)	8,760	0.85	710	18% (5.1%)
15-minute	19.7% (Obs.)	35,040	0.95	898	17% (2.7%)
1-minute	45% (Assumed)	525,600	0.99	2,641	22% (1.6%)

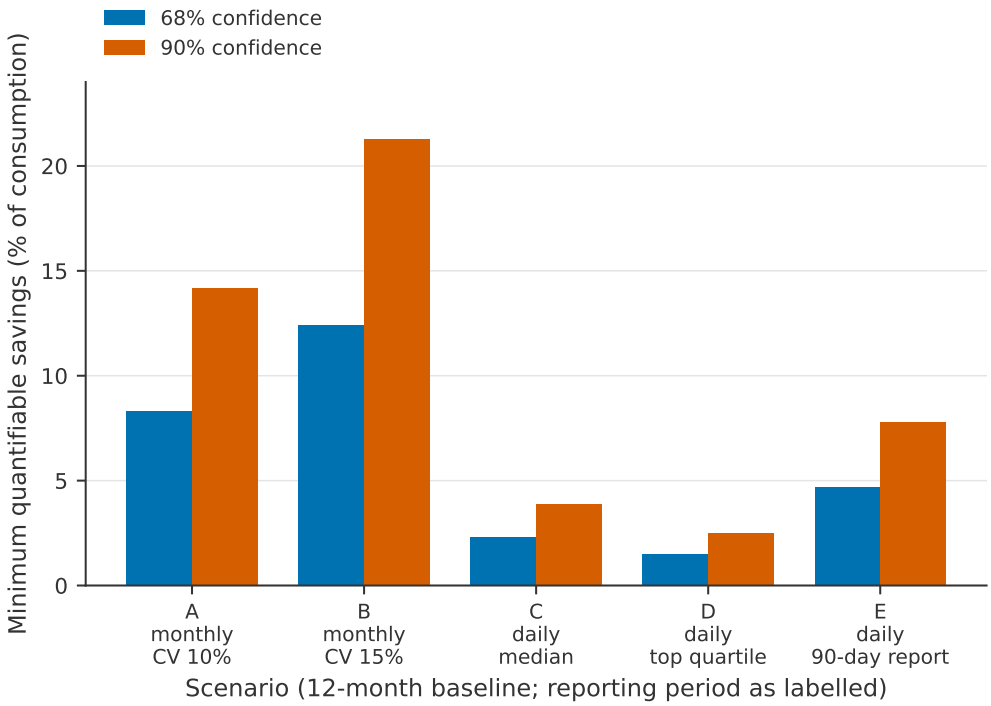


Figure 3: Minimum quantifiable savings by scenario and confidence level (from Table 5). The contrast between the monthly and daily regimes is the paper's central quantitative result.

aggregation of hourly data sits on the flat part of the curve), and a utility offering 1-minute data has not thereby made 2 percent savings routinely provable.

5.5 Detection is not quantification is not attribution

Three claims must not be conflated. Detection (the interval excludes zero) is achievable for savings down to about 1 to 2 percent at favourable sites with daily data and long reporting periods. Quantification (FSU at most 50 percent at 68 percent confidence) is the standard a payment, compliance claim, or published figure should meet; at a median building it requires savings of at least about 2.3 percent (68 percent) or 3.9 percent (90 percent) with a full year of daily data on both sides. Attribution (the intervention caused the reduction) is established by no uncertainty formula. The dominant threat is non-routine change: in most buildings and most years, the largest source of year-to-year change in energy use is neither efficiency measures nor weather but changes in operation and occupant behaviour (Granderson et al. 2015). A measured 4 percent reduction coincident with an unrecorded occupancy drop is detected, quantified, and wrongly attributed. The defences are procedural: non-routine-event screening and adjustment under IPMVP, comparison groups, residual monitoring to date the change, and measure-level corroboration.

There are conditions under which noise cannot be beaten down with any amount of data. Structural breaks in the unexplained component (tenant turnover, production change, schedule change, added or removed loads) break the counterfactual itself; finer data only measures the break more precisely. Buildings whose unexplained variance exceeds about 50 percent have no model-based remedy; the remedy is measurement (occupancy, production, submetering), not algorithms. Systematic bias does not shrink with sample size: a model with out-of-sample bias of -3 percent, well within the

observed interquartile range, mis-states every savings figure by 3 points at any duration or resolution. Extrapolation beyond the baseline envelope (a short baseline missing a season; unprecedented reporting-period weather) silently converts model error into apparent savings. And the FSU framework conditions on one fitted model: the spread across defensible models and baseline windows is an additional uncertainty layer that matters at small savings fractions. Climate and building type shift the numbers at the margin (in the 537-building study, Washington DC buildings' error distributions were notably larger than California's, attributed to more extreme seasonal temperature variation) but do not change the regime conclusions.

5.6 Programme gates and practitioner rules

The thresholds of Sections 5.3 and 5.4 are not only a literature result; in computed form they are the test current programmes enforce. The familiar rule that savings must exceed about 10 percent to be separable from noise descends from IPMVP's 2002-era guidance. The 2016 Core Concepts narrowed it: the 10 percent figure attaches to monthly data, and the rule of thumb for Option C on frequent data is about 5 percent (IPMVP, *M&V Issues and Examples*, 2017; Reconstructed-B). Programme rules carried the older form longer: California's NMEC Rulebook, in its 2020 version, still set a site-level aim of at least 10 percent of annual consumption within the metered boundary, with projects below that required to document how savings would be distinguished from normal variation. Version 2.1 of the Rulebook (2025) replaces the percentage floor with a computed per-site gate: fractional savings uncertainty of at most 50 percent of savings at 90 percent confidence, evaluated with the Guideline 14 forms (equations B27 and B33), the autocorrelation-corrected form applying where residuals are correlated; the hourly model need not pass if a model at another granularity does, and the forecast savings fraction used in the calculation is capped at 20 percent. The same revision eliminated R^2 as an eligibility criterion, expressly so that savings above or below 10 percent can qualify on model quality (Observed-A for the gate; the stated rationale is Observed-B, via the CPUC update record). Baseline screening follows the LBNL technical guidance the Rulebook incorporates (CV(RMSE) below 25 percent, NMBE within ± 0.5 percent, R^2 above 0.7), applied as a population screen and explicitly not as pass/fail criteria (Granderson et al. 2019; Observed-A).

An independent practitioner portfolio lands on the same floor this section computes. Kelly and Sinnamon (2020), analysing ComEd's Virtual Commissioning programme across 192 non-residential sites, found savings below about 3 percent not reliably detectable; hourly models slightly better than daily models at savings of 4 to 6 percent, and daily models equal at 7 percent and above; and about one third of the 84 sites with savings under 10 percent failed the programme's fitness criteria (NMBE below 0.5 percent, CV(RMSE) below 25 percent, FSU below 50 percent at 68 percent confidence). They add the caution that hourly FSU is likely understated, because the standard calculation understates uncertainty for hourly models (Observed-B). Set against Table 5, the agreement is close: detection at a median daily-data building from about 2 percent, quantification of 5 percent at both confidence levels, and no support for routine claims below 3 percent.

The Bonneville Power Administration's verified position is that no minimum savings threshold exists. Its *Regression for M&V Reference Guide* sets no floor in either its 2012 original or version 3.0 (2024): detectability is computed per project, expected savings must exceed twice the model's standard error, and where expected fractional uncertainty exceeds about ± 50 percent a different protocol is more appropriate. The guide notes that uncomfortably high relative precision is common for projects expected to save less than 5 percent, and its prescribed response is end-use submetering, which shrinks the measurement boundary rather than rejecting the project (Observed-A). BPA's Strategic Energy Management guide (v3.1, 2024) states the corresponding fitness table for operational savings:

net determination bias error below 0.5 percent; CV(RMSE) below 20 percent for daily models, 10 percent weekly, and 5 percent monthly; R^2 above 0.5; and predictive FSU below 50 percent at 68 percent confidence. Its uncertainty formula uses the same effective-sample-size correction as Section 5.2, $n' = n(1-\rho)/(1+\rho)$, and carries the caveat that the 1.26 coefficient may underpredict FSU for baseline periods longer than twelve months (Observed-A).

None of these gates lets R^2 do any work, and Stetz (EVO *M&V Focus*, Issue 5) shows why from the Guideline 14 algebra: uncertainty is driven by CV(RMSE) and the savings fraction, not by R^2. His cases are paired counterexamples: mosque sites with average R^2 above 0.9 still carried about 30 percent average uncertainty, and a site at a 9 percent savings fraction produced 81 percent uncertainty, so Option C was rejected for that site alone; conversely, a Riyadh office and data centre with R^2 of only 0.38 passed at 20 percent uncertainty (68 percent confidence), because CV(RMSE) was 0.16, the savings fraction was 30 percent, and the baseline ran 46 months. The same article states Section 5.3's aggregation arithmetic in practitioner form: site uncertainties combine in quadrature while savings add, so a bundle of sites met a criterion (10 percent savings at 90 percent confidence) that no single site could meet (Observed-B).

Not every programme computes the floor. Ontario's Energy Performance Program is a live pay-for-performance counterexample with a hard one: each facility must achieve at least 5 percent total electricity savings by the end of the second pay-for-performance period or it may be removed, eligibility requires 12 months of hourly interval data per facility, and baseline adjustments follow the EVO non-routine-events guide (EVO 10400-1:2020) under a materiality rule that an adjustment event should represent at least 0.5 percent of baseline consumption, namely 10 percent of the 5 percent threshold, be apparent on the hourly profile, and be independently verifiable (Observed-A). The programme's PY2023 evaluation found 77 percent of evaluated facilities reporting less than 5 percent of baseline savings in their first performance year (Observed-B). The floor there functions as a participation requirement rather than a detectability claim, and realised savings at real facilities cluster below it initially, which is this section's statistical point observed in programme data.

At building level, one further distinction matters: making a change visible is not the same act as proving it. The average-week hourly profile, the time-of-week view that time-of-week-and-temperature models produce, can make an operational intervention legible: in the ComEd work, a lighting schedule change (overnight shutdown combined with a delayed morning start) appears directly in the profile, in the hours the measure acts on. Programme practice uses the same instrument quantitatively: cumulative-sum charting of model residuals in BPA's SEM protocol dates slope changes against the programme's opportunity register, so an inflection can set a measure's effective date, and the absence of unexplained inflections is taken as evidence that no significant non-routine event occurred. Such visibility is evidence for the attribution narrative of Section 5.5, not a substitute for the uncertainty calculation: a shape change can be plain to the eye while the cumulative claim still fails the FSU gate, and the gate is what a payment should turn on. Comparison and control buildings, often proposed as the single-site attribution remedy, are in documented practice a population-level instrument (population-level NMEC adjusts for exogenous effects with comparison groups specified in the programme M&V plan); no programme practice of matched control buildings for a single site was found in the sources reviewed, and single-site attribution rests instead on non-routine-event logs, adjustment workflows, and load-shape evidence.

Three cautions close the practitioner record, and all three point the same way. The EVO Snapshot on Advanced M&V states that current FSU calculations are not reliable with hourly or daily data and tend to underestimate uncertainty. The LBNL methods literature demonstrates the understatement empirically and attributes its hourly severity to stronger residual autocorrelation (Section 5.2). BPA's SEM guide warns that even the corrected formula's 1.26 coefficient may underpredict FSU

for baselines beyond twelve months. This paper therefore reports autocorrelation-corrected figures throughout this section and treats them as floors rather than central estimates. Two limits of the practitioner record itself are recorded rather than smoothed over: EVO's announced standalone application guide on advanced meter-based M&V methods could not be located as released, and its contents are Unknown; and EVO's *Uncertainty Assessment for IPMVP* (10100-1:2018) is paywalled, so its equations are used here at Reconstructed-B through secondary reproduction. A final caution concerns independence: the strands are not separate witnesses, because LBNL staff (Granderson and colleagues) authored the portfolio studies of Section 5.1, the NMEC technical guidance California incorporates, and contributed to the EVO Snapshot, so the practitioner rules of thumb and the portfolio statistics are the same research community speaking at two levels of rigour, and their agreement is a consistency check on the floor computed here, not independent replication of it.

6 Joining the two halves

Table 7 crosses the documented data regimes of Sections 3 and 4 with the statistical requirements of Section 5. The 12-month baseline that the daily-data scenarios require is obtainable through documented customer channels in most of the EU (as a right, with national depth varying), in the best-documented North American markets (12 to 24 months), and in Australia (multi-year, against a 7-year retention rule). It is not assured where access is bilateral or paid, and it is not verifiable at all where retention is unpublished, because a channel that exists today cannot promise a baseline that was recorded before the customer asked. The single most consequential institutional fact for savings verification is therefore not resolution; it is that retention is undocumented in most jurisdictions, which converts every long-baseline design into a negotiation.

Table 7: Data regimes against claim sizes (daily-aggregated interval data, 12-month baseline and reporting, median building; from Table 5)

Regime	2% claim	3–5% claim	8–10% claim
15/30-min data, ≥12 mo history obtainable (EU right; TX, CA, NY, IL, Ontario; AU; NZ from 2027)	Detectable; not quantifiable at an ordinary site	Quantifiable (3% at 68%; 5% at 90%)	Quantifiable, robust
Hourly data only, ≥12 mo history	Same as above after daily aggregation	Same	Same
Monthly billing data only	Not detectable	5% detectable at 68% only (good model)	10% quantifiable at 68% (good model)
History < baseline need, or retention unknown	Baseline truncated: accuracy degrades (Table 3, training-length rows)	Weakened proportionally	Usually still supportable
1-min data, any history	No material gain over 15-min	No material gain	No material gain

The practical consequences follow. A demand-management or efficiency provider selling single-digit percentage savings at large C&I sites can, in the documented regimes, build a defensible verification design on utility data alone: 15 or 30-minute records, 12 months of baseline obtained at contract start (before retention becomes the counterparty's question), daily-aggregated whole-building models, and claims sized at 5 percent and above for quantification, 3 percent at moderate confidence, 2 percent as

detection only. The same provider cannot defensibly verify those claims from monthly bills, cannot improve them materially by demanding minute-level data, and should treat any jurisdiction where retention is unpublished as a data risk to be retired contractually (by exporting the baseline at engagement start) rather than discovered later.

7 What would change these conclusions

The institutional conclusions would change if (i) major jurisdictions publish retention schedules materially different from the pattern found here (for example, a German or French rule fixing short retention would lower the coverage estimates of Section 4; published multi-year schedules would raise them); (ii) the European implementing framework or national hub rules fix a harmonised history depth, converting the EU right of access into a uniform quantity; or (iii) ERCOT's post-2025 settlement design and Ontario's mandated Green Button history depth, both flagged as unverified in this study, resolve to finer or deeper standards than the secondary evidence suggests. The statistical conclusions would change if (iv) portfolio-scale out-of-sample studies at 1-minute resolution demonstrated materially lower daily-aggregated error than the 15-minute studies found here, which would require the day-scale error component to be smaller than the 537-building evidence indicates; or (v) a model class demonstrated, out of sample and at portfolio scale, median daily CV(RMSE) well below the 12.9 percent found across the ten models of Granderson et al. (2016), which would lower every threshold in Table 5 proportionally. The claim that detection is not attribution would change only with evidence that operational change is smaller, relative to measure effects, than Granderson et al. (2015) found. Absent such evidence, the conclusions stand.

8 Conclusion

Utilities, market operators, and data hubs do retain sufficiently granular historical consumption data for large C&I customers in most of the documented world: 15-minute records across the EU and the best-documented North American markets, 30-minute records across much of Asia-Pacific, 5-minute records in a minority of markets. Customers can realistically obtain that data, at 12 to 24 months of history, in the jurisdictions that have made access a right or a market rule, which cover roughly half of the documented sample. The weak layer is retention: unpublished in most jurisdictions, it is the point at which an otherwise sound verification design can silently fail.

Statistically, the data that realistically exist are sufficient for the claims that matter most and insufficient for the claims that are most often marketed. Daily or hourly interval data with a year on both sides quantify savings of 5 percent and above at a typical site, quantify 3 percent at moderate confidence, and detect 2 percent without quantifying it. Monthly billing data support none of these claims at a single site. Finer-than-hourly data do not rescue small claims, because the noise that binds is the building's own operational variability at the scale of days and seasons. The honest summary for any party proposing to prove small savings from meter data is: the meter keeps enough; the building is the noise; and the difference between a defensible claim and an indefensible one is the size of the claim relative to the building's demonstrated predictability, not the resolution of the record.

References

ASHRAE (2014). *Guideline 14-2014: Measurement of Energy, Demand, and Water Savings.* American Society of Heating, Refrigerating and Air-Conditioning Engineers.

Bonneville Power Administration (2020). *C&I SEM M&V Reference Guide*, v3.0.

Bonneville Power Administration (2024). *C&I Strategic Energy Management (SEM) M&V Reference Guide*, v3.1.

Bonneville Power Administration (2024). *Regression for M&V: Reference Guide*, v3.0.

California Public Utilities Commission (2025). *Rulebook for Programs and Projects Based on Normalized Metered Energy Consumption (NMEC)*, version 2.1.

CalTRACK Methods (OpenEEmeter/OpenDSM). Site-level fractional savings uncertainty, Section 4.3.2.4.

Directive (EU) 2019/944 on common rules for the internal market for electricity, Articles 20 and 23.

Implementing Regulation (EU) 2023/1162 on interoperability requirements and non-discriminatory and transparent procedures for access to metering and consumption data.

Regulation (EU) 2017/2195 (Electricity Balancing Guideline), Article 53 (15-minute imbalance settlement period).

Efficiency Valuation Organization (2020). *IPMVP Application Guide on Non-Routine Events and Adjustments*, EVO 10400-1:2020.

FERC (2016). Order No. 825: Settlement Intervals and Shortage Pricing in Markets Operated by Regional Transmission Organizations and Independent System Operators.

Granderson, J., Price, P.N. (2014). Development and application of a statistical methodology to evaluate the predictive accuracy of building energy baseline models. Lawrence Berkeley National Laboratory.

Granderson, J., Price, P.N., Jump, D., Addy, N., Sohn, M.D. (2015). Automated measurement and verification: Performance of public domain whole-building electric baseline models. *Applied Energy* 144, 106–133.

Granderson, J., Touzani, S., Custodio, C., Sohn, M.D., Jump, D., Fernandes, S. (2016). Accuracy of automated measurement and verification (M&V) techniques for energy savings in commercial buildings. *Applied Energy* 173, 296–308.

Granderson, J., Touzani, S., Fernandes, S., Taylor, C. (2017). Application of automated measurement and verification to utility energy efficiency program data. *Energy and Buildings* 142, 191–199.

Granderson, J. et al. (2019). LBNL Guidance on M&V 2.0 Savings Claims (draft guidance). Lawrence Berkeley National Laboratory.

Granderson, J. et al. (2019). *Site-Level NMEC Technical Guidance: Program M&V Plans Utilizing Normalized Metered Energy Consumption Savings Estimation*, version 2.0. Lawrence Berkeley National Laboratory, for the California Public Utilities Commission Energy Division.

Independent Electricity System Operator / Save on Energy (2025). *Energy Performance Program Participant Agreement*, version 2.0.

Kelly, K., Sinnamon, D. (2020). Detecting savings under 10% using IPMVP Option C. *EVO M&V Focus* Issue 7, October 2020.

Mathieu, J.L., Price, P.N., Kiliccote, S., Piette, M.A. (2011). Quantifying changes in building electricity use, with application to demand response. *IEEE Transactions on Smart Grid* 2(3), 507–518.

Reddy, T.A. et al. (1997). Baselining methodology for facility-level monthly energy use. *ASHRAE Transactions* 103(2).

Stetz, M. Why R2 doesn't matter. *EVO M&V Focus* Issue 5.

Touzani, S., Granderson, J., Fernandes, S. (2018). Gradient boosting machine for modeling the energy consumption of commercial buildings. *Energy and Buildings* 158, 1533–1543.

US DOE Federal Energy Management Program (2024). *M&V Guidelines: Measurement and Verification for Performance-Based Contracts*, Version 5.0.

Walter, T., Price, P.N., Sohn, M.D. (2014). Uncertainty estimation improves energy measurement and verification procedures. *Applied Energy* 130, 230–236.

Webster, L. et al. (2020). *IPMVP's Snapshot on Advanced Measurement and Verification.* Efficiency Valuation Organization.

Institutional sources for Section 3 (regulator orders, market rules, metering codes, and utility documentation, each with locator and retrieval date of 16 October 2026) are catalogued in full in the paper's research files: PUCT Rule 25.130 and Smart Meter Texas documentation; CPUC Decision 11-07-056 and PG&E Rule 22; NYPSC Case 20-M-0082 Data Access Framework and Con Edison Green Button Connect documentation; ICC 17-0123; PJM Manual 25 and ISO New England settlement documentation; CAISO tariff (metering and settlement-quality meter data provisions); IESO Market Rules Chapter 6; Ontario Regulation 633/21; Hydro-Québec tariffed load-profile services; Danish Executive Order 477/2023 and DataHub documentation; German PTB-A 50.8; UK Smart Energy Code and government business guidance on DCC data; AEMO metering and Metering Data Provision Procedures and the Australian Consumer Data Right (energy); New Zealand Customer and Product Data Act 2025 designation materials; CCEE metering and SCDE documentation (Brazil); KEPCO AMI documentation (South Korea); CEA/CERC metering regulations (India); Eskom Megaflex tariff documentation (South Africa); SP Group/MSSL documentation (Singapore).

A Jurisdiction tables (condensed)

The tables below condense the regional evidence files. "Unknown" means no documented value was found in primary or authoritative sources in this study's retrieval pass; it does not mean the value is zero or the practice absent.

North America

System	Recorded	Settlement	Obtainable history	Retention (documented)
ERCOT / Texas TDSPs	15 min	15 min	$\leq$36 mo authorised; $\sim$12 mo download	Unknown
CAISO / California IOUs	5/15/60 elected	5 min	12 mo (PG&E Rule 22)	SQMD $\geq$3 yr (DER aggregations)
PJM	hourly (load)	5 min (derived)	set by state/EDC	Unknown
ISO New England	hourly (load)	5 min (from 2017)	set by state	Unknown
NYISO / Con Edison	hourly wholesale; 5 min AMI	hourly load	24 mo cap	Unknown
Illinois / ComEd	15 min	PJM 5 min	12–24 mo; sharing indefinite	Unknown
Duke NC	15 min	n/a	12–24 mo	Unknown
Duke Ohio	15 min	PJM 5 min	12 mo (aggregators)	Unknown
Duke Florida	15 min (portal)	n/a	Unknown; no GBC/API	Unknown
Ontario wholesale (IESO)	5 min	hourly price (non-dispatchable)	participant channels	Unknown

System	Recorded	Settlement	Obtainable history	Retention (documented)
Ontario distribution	hourly	TOU from repository	depth unverified	Unknown
BC Hydro	Unknown (large C&I)	n/a	hourly portal	Unknown
Hydro-Québec	Unknown	n/a	paid load-profile services	≥4 yr demonstrated

Europe

Jurisdiction	Recorded	Transmission	Retention (documented)	Obtainable
EU framework	15 min	n/a	no EU-wide period	statutory right (Art. 20)
Germany	15 min (5-min measurement possible)	market flows	Unknown	supplier/operator Lastgang
France	unverified	daily (Linky)	Unknown	Enedis space/services
Great Britain	30 min	daily/periodic	13 mo HH meter-resident (DCC)	DCC; SEC consent route
Denmark	15 min	to DataHub	messages ≥3 yr; personal data 5 yr	hub portal
Netherlands	15 min	via metering cos.	Unknown	contracted metering company
Spain	Unknown	Unknown	Unknown	hub exists; depth Unknown
Italy	Unknown	Unknown	Unknown	Unknown
Sweden	15 min (from Nov 2023)	settlement chain	Unknown	portal
Norway	15-min hub-ready 2023	Elhub to eSett	6 yr (attribution unverified)	hub portal
Austria	15 min (opt-in) / daily	daily	60 days on-meter	portal at configured resolution
Poland	Unknown	into CSIRE	Unknown	CSIRE, free of charge

Rest of world

Jurisdiction	Settlement	Recorded	Customer can obtain	Retention (documented)
Australia (NEM)	5 min	5 min	5-min via request; CDR (<5 GWh/yr)	≥7 yr (MDP)
Philippines (WESM)	5 min	5 min	via retailer/statements	Unknown
Brazil (CCEE)	hourly	5 min recorded	5-min via SCDE agent access	Unknown

Jurisdiction	Settlement	Recorded	Customer can obtain	Retention (documented)
China (spot prov.)	15 min	15 min	via trading centre/retailer	Unknown
South Korea	n/a	15 min (AMI)	portal; MyData (scope unclear)	Unknown
India	15-min blocks	15 min	load survey on request	Unknown
UAE (Dubai)	n/a	15 min	app history; granularity unspecified	Unknown
New Zealand	30 min	30 min	retailer; CDR from Jul 2027 (2-yr)	≥2 yr implied
Singapore	30 min	30 min	via retailer/MSSL	Unknown
Japan	30 min	30 min	via retailer/T&D	Unknown
Vietnam	30 min	30 min	undocumented	Unknown
South Africa	n/a	30 min	bilateral (Eskom)	Unknown
Chile	hourly	hourly	bilateral	Unknown
Saudi Arabia	n/a	smart fleet (spec unpublished)	portal views; no documented right	Unknown
Pakistan	hourly (CTBCM design); monthly billing	30 min (AMI spec; configurable to 1 min)	no documented interval product	Unknown
Malaysia	monthly billing	30 min	up to 30-min via myTNB portal; depth undocumented	Unknown
Bangladesh	monthly billing	30 min (load survey spec)	portal access specified in DPDC AMI tender; live status unverified	in meter: ≥90 days load survey, 12 mo energy/demand; MDM Unknown
Sri Lanka	monthly billing	15 min (LECO, residential only); C&I Unknown	no documented C&I access	Unknown
Nepal	monthly TOD billing	TOD block registers; width Unknown	app view (smart meters); ad hoc summaries	Unknown
Indonesia	monthly billing	30 min (Bali pilot); production AMR Unknown	described as accessible; granularity Unknown	Unknown
Thailand	monthly billing	Unknown (AMR for 400+ largest C&I deferred)	Unknown	Unknown
Türkiye	hourly	hourly (OSOS)	participant settlement data; general right Unknown	storage duty; period Unknown
UAE (Abu Dhabi)	PWPA settlement	hourly at settlement points; customer demand 5 min–1 h programmable	app usage view; granularity Unknown	≥40 days at outstation; customer data Unknown
Qatar	monthly billing	Unknown (AMI at scale)	monitoring claimed; granularity Unknown	Unknown
Oman	monthly billing	hourly (AMR, >150 MWh/yr)	not documented	Unknown

Jurisdiction	Settlement	Recorded	Customer can obtain	Retention (documented)
Israel	via suppliers; granularity Unknown	Unknown (smart fleet at scale)	switching enabled; self-service product Unknown	Unknown
Egypt	monthly/prepaid	Unknown (prepaid replacement at scale)	not documented	Unknown
Kenya	monthly billing	Unknown (smart metering for large power)	app bills; interval product not documented	Unknown
Nigeria	monthly billing	30 min (MD storage in meter)	monthly MD figure; no documented right	6 mo in meter; central Unknown
Argentina	hourly spot	15 min (SMEC; grade B)	agent settlement data via CAMMESA; granularity Unknown	Unknown
Colombia	hourly	hourly frontier reporting	agent/ASIC channels; end-customer right Unknown	Unknown

Production note

This working paper was produced by the Kilambi Research pipeline: six parallel evidence streams compiled from primary and peer-reviewed sources retrieved on 16 October 2026, namely the four original streams (three regional documentation streams and one statistical literature stream) and two supplementary streams of 16 October 2026 (a seventeen-jurisdiction extension of the rest-of-world coverage and a practitioner-literature stream), followed by synthesis, computation of all threshold tables from the stated formulas, and drafting with AI assistance under the author's direction. Every quantitative claim is graded by evidence type and quality in the underlying research files. The paper is published as KR-WP-2026-30 on research.kilambi.com.